Mehul Khanna

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Summary

Final-year B.E. Robotics & AI student (TIET, 2027) seeking a Software Engineer role. Builds and tests Python applications end to end: REST APIs, SQL databases, automated tests in CI, Docker and Linux. Former vulnerability management intern. Open to development, testing and production support roles, and to relocation.

Education

Thapar Institute of Engineering & Technology (TIET) **2023 – 2027 (Expected)**
Bachelor of Engineering (B.E.), Robotics & Artificial Intelligence *Patiala, India*
Relevant Coursework: Programming for Problem Solving (C), AI for Engineers, Optimization Methods, Numerical Analysis, Mechatronic Systems

Experience

Vulnerability Management Intern **Jun 2025 – Aug 2025**
CodeAsia *Experience Letter* *Internship*
- Assisted **vulnerability assessments** of authorised web applications, mobile apps and network assets; verified each finding before reporting.
- Rated **severity** with the company's risk process, prioritised fixes, and tracked **remediation** with owners until confirmed.
- Wrote **Python scripts** to automate routine vulnerability checks and reporting, and prepared assessment summaries.

Technical Skills

Languages: Python, C++, C, SQL, Bash | **Core CS:** OOP, Operating Systems, Computer Networks
Backend & Databases: Flask, FastAPI, REST APIs (JSON), PostgreSQL, SQLite, SQLAlchemy Core, schema design, indexing
Testing & DevOps: pytest, unit testing, integration testing, test cases, test automation, debugging, Git, GitHub Actions (CI/CD), Docker, Linux, log analysis, troubleshooting
Security: vulnerability assessment, risk rating, remediation tracking, OWASP Top 10, responsible disclosure
AI/ML: scikit-learn, OpenCV, NumPy, pandas, LLM integration, Retrieval-Augmented Generation (RAG), llama.cpp

Projects

OpsPilot — Incident Triage & Support Desk | *Python, Flask, PostgreSQL, pytest, Docker, GitHub Actions* GitHub
- Built a production-support app (REST API + dashboard): a **log analyzer** detects per-service error spikes with a sliding window and opens incidents with SLA deadlines and an enforced lifecycle.
- Designed a normalized **PostgreSQL** schema (foreign keys, constraints, partial indexes) with raw SQL **JOIN**/GROUP BY reports for SLA breaches and mean time to resolve.
- Added AI triage (BM25 runbook **RAG** + LLM, rule-based fallback) that engineers approve or reject, with an audit trail; 120+ **pytest** tests in **GitHub Actions** on SQLite and PostgreSQL.

Praxis — Linux Desktop with Local-First AI | *Python, FastAPI, SQLite FTS5, llama.cpp, Qt/QML, Arch Linux* GitHub
- Built an Arch-based Linux distribution (bootable ISO, graphical installer, Qt/QML shell) around a **FastAPI** backend.
- Wrote an LLM gateway (local llama.cpp + cloud) with SQLite FTS5 memory retrieval, and a default-deny layer that sorts AI-generated shell commands into auto, confirm or forbidden.

Semantic Frame Retention | *Python, OpenCV, scikit-learn, Flask* GitHub
- Built a video pipeline that keeps only important frames using novelty scores and an ML decision gate, explains each decision, and is evaluated with ablations.

Also: Robotic Arm Pick & Place — ROS 2 simulation of a URDF arm with a Python controller, visualised in RViz2.

Leadership

Technical Head & Core Board Member
OWASP TIET Student Chapter *Patiala, India*
- Led the technical team running security workshops and Capture-the-Flag (CTF) events for **150–200 participants**; coordinated the Spectre CTF (Jan 2025), recognised with a certificate of appreciation.

Achievements & Certifications

- **NASA Vulnerability Disclosure Program Hall of Fame** — responsible disclosure of a security vulnerability
- **Runner-Up**, Hack4Sustainability 2023 | **Top 28 of 200+ teams**, Smart India Hackathon 2023 (TIET round) | **OCI AI Foundations Associate** (Oracle, 2025)